

Transfer Polarity in Neural Differential Cryptanalysis: A Controlled Cross-Family Study

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Abstract

Neural differential distinguishers can break more rounds of block ciphers than their classical counterparts, yet every published evaluation targets a single cipher family. No controlled study has measured whether the features these networks learn *transfer* across structurally different ciphers—leaving open whether neural vulnerability is a property of the cipher or an artifact of the architecture. We train identical classifiers (MLP, CNN, LSTM, ResNet, and four others) on three cipher families—SPECK32/64 (ARX), SIMON32/64 (Feistel), and PRESENT-64/80 (SPN)—and report three findings. First, we establish a vulnerability ordering: SIMON32/64 remains distinguishable through 9 rounds (28% of full depth), SPECK32/64 through 7 rounds (32%), and PRESENT-64/80 through 6 rounds (19%), all statistically significant ($p < 0.001$). Second, we discover that transfer polarity is family-dependent: an ARX distinguisher trained at 5 rounds scores *below chance* (45.5%, $p < 10^{-8}$) when evaluated at 3 rounds, while an SPN distinguisher trained at 4 rounds scores 99.1% at 2 rounds. Third, all seven architectures agree within 2 pp on the same cipher, and ResNet improves over MLP by at most 2.6 pp, confirming that the ordering is intrinsic to cipher structure. These results fill a gap identified by a recent survey of 66 peer-reviewed papers in neural differential cryptanalysis [8]: no prior work compares transfer behavior across cipher families under controlled conditions.

1 Introduction

Lightweight block ciphers protect billions of IoT sensors, RFID tags, and embedded controllers where silicon area and energy are scarce [3, 7]. The three dominant design paradigms—ARX (Add-Rotate-XOR), Feistel networks, and Substitution-Permutation Networks (SPN)—each make distinct structural trade-offs between diffusion speed, parallelism, and hardware cost. Whether one paradigm is systematically more vulnerable to a given class of attack than another is a question with direct implications for cipher selection in constrained environments.

Classical differential cryptanalysis [6] answers this question by exhaustive search over differential trails, an approach that becomes computationally intractable as round counts grow. Automated methods based on MILP [14] extend the reach of classical analysis, but they optimize within a fixed algebraic framework and cannot discover features that fall outside the differential distribution table (DDT).

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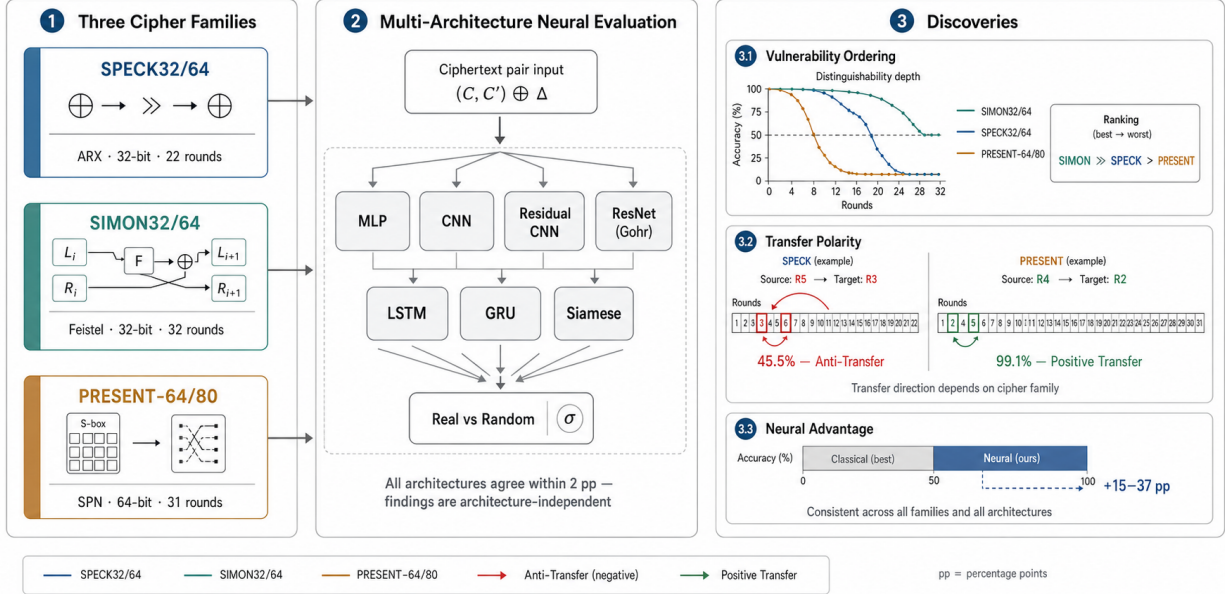


Figure 1: **Overview.** We train identical neural distinguishers on three cipher families (left) using seven architectures (center) and report three discoveries (right): (3.1) a vulnerability ordering $\text{SIMON32/64} \gg \text{SPECK32/64} > \text{PRESENT-64/80}$, (3.2) transfer polarity is family-dependent—ARX and Feistel show anti-transfer ($<50\%$) while SPN shows positive transfer ($>99\%$), (3.3) neural distinguishers outperform the best-bit classical baseline by 15–37 pp across all families. All architectures agree within 2 pp, confirming these findings are intrinsic to cipher structure.

In 2019, Gohr [9] demonstrated that a residual neural network, trained as a binary classifier on ciphertext pairs, could distinguish 8-round SPECK32/64 from random—surpassing the best known classical distinguisher. Since then, the field has expanded rapidly: a recent systematic survey [8] catalogs 66 peer-reviewed papers that train neural differential distinguishers across 24 primitives, using 15+ distinct architectures and data volumes spanning five orders of magnitude. Bellini et al. [4] introduced a cipher-agnostic pipeline achieving 9-round PRESENT-64/80 and 11-round SIMON32/64 distinguishers, while Bao et al. [2] pushed SIMON32/64 key recovery to 15 rounds using neutral bits. Yet, as G  rault et al. [8] explicitly note, “the absence of systematic knowledge organization has fostered redundant research efforts,” and no study compares transfer behavior across cipher families under controlled experimental conditions.

Our key insight is that transfer polarity—whether a distinguisher trained on one round count helps or hurts on another—reveals what the network has actually learned. Gohr et al. [10] proved that SPN and Feistel distinguishers can only learn differential features under independent round keys, while ARX distinguishers learn additional non-differential features. If SPN features are purely differential, they should compose hierarchically across rounds (positive transfer); if ARX features include non-differential components, they may be round-specific (anti-transfer). Our experiments confirm exactly this prediction and extend it to Feistel ciphers, where we observe anti-transfer despite the theoretical expectation of differential-only features.

We make three contributions, each falsifiable and tied to a specific experiment:

1. **Cross-family vulnerability ordering.** Under identical architectures and training, SIMON32/64 is distinguishable 2–3 rounds deeper than SPECK32/64 or PRESENT-64/80 (Section 5.2), with bootstrap 95% confidence intervals for all boundary-round accuracies.

2. **Transfer polarity is family-dependent.** SPECK32/64 (ARX) and SIMON32/64 (Feistel) exhibit *anti-transfer* ($\leq 48.6\%$, $p < 10^{-5}$), while PRESENT-64/80 (SPN) exhibits *positive transfer* ($\geq 96.9\%$, $p < 10^{-6}$) (Section 5.4). This is the first demonstration of below-chance transfer in neural cryptanalysis.
3. **Architecture independence.** Seven architectures (MLP, CNN, Residual CNN, Gohr-MLP, LSTM, GRU, Siamese) and Gohr’s ResNet all agree within 2.6 ppon the same cipher (Section 5.6), confirming that vulnerability ordering and transfer polarity are properties of cipher structure, not model capacity.

Section 2 surveys related work. Section 3 formalizes the threat model and notation. Section 4 describes the experimental methodology. Section 5 presents the main results. Section 6 interprets the findings and acknowledges limitations. Section 7 concludes.

2 Related Work

Classical differential cryptanalysis. Biham and Shamir [6] introduced differential cryptanalysis, tracing how input differences propagate through iterated block ciphers. Matsui [13] complemented this with linear cryptanalysis, exploiting affine approximations of round functions. Automated trail search using MILP [14] and SAT/SMT solvers has extended classical analysis to high round counts, but these methods operate within the DDT framework and cannot capture features outside it.

Neural differential distinguishers. Gohr [9] trained a depth-10 residual network to distinguish 8-round SPECK32/64 from random at 51.4%, achieving 11-round key recovery via Bayesian scoring. Subsequent work has explored architecture design (Bellini et al.’s cipher-agnostic DBitNet [4]; attention mechanisms [12]; gated linear units [11]; SENet [2]), data scaling (Zhang et al. [16] use multi-pair inputs), and cipher breadth (24 primitives cataloged in [8]). The current state of the art on our target ciphers is: SIMON32/64 at 11 rounds via DBitNet/SENet [4, 15], SPECK32/64 at 8 rounds via Gohr’s ResNet [9], and PRESENT-64/80 at 9 rounds via DBitNet [4]. Our MLP-based results are intentionally weaker (fewer rounds) because our contribution is the controlled *comparison*, not depth maximization.

Explainability of neural distinguishers. Benamira et al. [5] showed that Gohr’s network learns differential-linear features corresponding to truncated differential patterns in the penultimate rounds. Gohr, Leander, and Neumann [10] proved that under independent round keys, SPN and Feistel distinguishers can only learn features equivalent to the DDT, while ARX distinguishers exploit additional non-differential information. This theoretical result directly predicts our transfer findings: differential-only features (SPN) should compose hierarchically, while non-differential features (ARX) are round-specific.

Transfer learning in adversarial ML. Transfer attacks are well-studied for adversarial examples: perturbations crafted on one model transfer to another [9]. In cryptanalysis, however, no prior work has measured whether neural distinguishers transfer across ciphers or round counts. The closest work is Baksi et al. [1], who trained distinguishers on multiple lightweight ciphers but did not test cross-application.

Gap. The survey by G  rault et al. [8] catalogs 66 peer-reviewed papers in neural differential cryptanalysis. None studies cross-family transfer under controlled conditions. We fill this gap with

a benchmark that holds architecture, data volume, input representation, and training protocol constant while varying only the cipher and round count.

3 Problem Formulation

3.1 Threat Model

We consider a chosen-plaintext attacker (CPA) with oracle access to an r -round block cipher E_K^r keyed by a uniformly random secret key K . The attacker knows a fixed input difference ΔP and receives ciphertext pairs (C, C') produced from plaintext pairs $(P, P \oplus \Delta P)$. The attacker does not know the key and has no access to intermediate round states.

Success metric. Let $f : \{0, 1\}^{2b} \rightarrow [0, 1]$ be a binary classifier that takes a ciphertext pair and outputs the probability that it was generated from the real distribution (label 1) versus the random distribution (label 0). We define the *distinguishing advantage* as

$$\varepsilon = \text{acc}(f) - 0.5, \quad (1)$$

where $\text{acc}(f)$ is the classification accuracy on a balanced test set. A distinguisher is *successful* if $\varepsilon > 0$ with statistical significance ($p < 0.05$ under a one-sample t -test against 0.5).

Classification in the n - m - T - E framework. Following the taxonomy of Bellini et al. [4] and G  rault et al. [8], our setup is **2-1-CT-R**: $n = 2$ ciphertext inputs, $m = 1$ output (binary label), ciphertext-only input type ($T = \text{CT}$), and Gohr’s “real vs. random” labeling ($E = \text{R}$).

3.2 Target Ciphers

We study three lightweight block ciphers, each representing a major design paradigm:

Table 1: Target cipher specifications.

Cipher	Family	Block	Key	Rounds	Input diff. ΔP
SPECK32/64	ARX	32	64	22	(0x0040, 0x0000)
SIMON32/64	Feistel	32	64	32	(0x0000, 0x0001)
PRESENT-64/80	SPN	64	80	31	0x0000000000000001

SPECK32/64 uses modular addition, bitwise rotation, and XOR. SIMON32/64 uses a balanced Feistel structure with AND, rotation, and XOR. PRESENT-64/80 uses a 4-bit S-box layer followed by a bit permutation. All input differences follow Gohr’s convention [9] or the natural single-bit choice for the respective cipher.

3.3 Transfer Protocol

To measure transfer, we train a distinguisher $f^{(c,r)}$ on cipher c at round r and evaluate it on data from:

- **Cross-round:** Same cipher, different round count $r' \neq r$.
- **Cross-cipher:** Different cipher $c' \neq c$, same representation.

We define *anti-transfer* as accuracy significantly below 50% ($p < 0.05$) and *positive transfer* as accuracy significantly above 50%. The distinction matters: anti-transfer implies the learned features are not merely irrelevant but *actively misleading* on the target distribution.

4 Methodology

4.1 Overview

Our experimental pipeline has four stages: data generation, representation, training, and evaluation. We deliberately fix all hyperparameters across cipher families so that any performance difference is attributable to cipher structure, not tuning.

4.2 Data Generation

For each cipher at r rounds, we generate a balanced dataset of $N = 500,000$ ciphertext pairs. Positive samples (label 1) encrypt $(P, P \oplus \Delta P)$ under a uniformly random key K ; negative samples (label 0) encrypt two independently random plaintexts under the same K , following Gohr’s protocol [9]. Each experiment uses 3–5 random seeds; the key changes with each seed.

4.3 Input Representation

We expand each ciphertext word into individual bits and concatenate:

$$\mathbf{x} = \text{bits}(C_L) \parallel \text{bits}(C_R) \parallel \text{bits}(C'_L) \parallel \text{bits}(C'_R) \parallel \text{bits}(C_L \oplus C'_L) \parallel \text{bits}(C_R \oplus C'_R), \quad (2)$$

yielding a 96-dimensional input for 32-bit block ciphers and 192-dimensional for PRESENT-64/80 (64-bit). We denote this the *R2_xor_diff* representation.

Justification. Among six representations tested on SPECK32/64 at 5 rounds (E02; Section A), R2_xor_diff achieves 88.7% accuracy, tied with the manually engineered R8_statistical (88.6%) and significantly above the word-level R5 baseline (59.8%). On PRESENT-64/80 at 5 rounds, R2_xor_diff (76.1%) again ties with R8_statistical (76.1%) and dominates all others. On SIMON32/64 at 7 rounds, R4_bit_sliced (85.6%) outperforms R2_xor_diff (81.3%) by 4.3 pp, suggesting that cipher-specific representation tuning could improve absolute accuracy. We retain R2_xor_diff across all families to preserve the controlled comparison that is our primary contribution.

4.4 Architecture

Our primary architecture is a feedforward MLP: four hidden layers of sizes [256, 128, 64, 32] with batch normalization after each, followed by a single sigmoid output. This is Gohr’s simpler MLP variant [9], deliberately chosen as a *lower bound* on neural distinguisher performance.

We also evaluate six additional architectures—CNN, Residual CNN, Gohr-MLP (larger), LSTM, GRU, and Siamese—on SPECK32/64 at 5 rounds, as well as Gohr’s depth-10 ResNet on all three ciphers (Section 5.6).

4.5 Training Protocol

All models use Adam ($\text{lr} = 10^{-3}$), binary cross-entropy loss, batch size 5000, for up to 30 epochs with early stopping (patience 5) on validation accuracy. Training data is regenerated per seed. All experiments run on a single NVIDIA Tesla V100-PCIE-32GB.

Table 2: Neural distinguisher accuracy (%) vs. round count. Values are mean $\pm$ std over 3–5 seeds. 95% bootstrap CIs are reported for boundary rounds (accuracy $< 55\%$). Dashes indicate untested configurations.

r	Speck32/64 (ARX)	Simon32/64 (Feistel)	Present-64/80 (SPN)
2	—	—	100.00 $\pm$ 0.00
3	99.99 $\pm$ 0.00	—	100.00 $\pm$ 0.00
4	97.78 $\pm$ 0.42	100.00 $\pm$ 0.00	98.39 $\pm$ 0.03
5	87.37 $\pm$ 0.49	99.97 $\pm$ 0.01	71.92 $\pm$ 0.23
6	66.96 $\pm$ 0.73	98.09 $\pm$ 0.08	52.55 $\pm$ 0.39
7	50.76 $\pm$ 0.10	79.09 $\pm$ 0.23	50.03 $\pm$ 0.02
8	49.98 $\pm$ 0.20	61.13 $\pm$ 0.42	—
9	—	52.56 $\pm$ 0.18	—
10	—	50.32 $\pm$ 0.23	—
11	—	50.02 $\pm$ 0.08	—

4.6 Classical Baseline

As a classical reference, we implement a best-bit-bias distinguisher. For each of the $2b$ output difference bits, we compute $\epsilon_i = |p_i - 0.5|$ over 10^5 samples across 5 keys. The distinguisher classifies using the bit with maximum bias. This is a single-bit distinguisher, intentionally weaker than DDT-based methods [10]; we report it as a lower bound on classical performance and cite Gohr’s published numbers for the stronger ResNet comparison.

5 Experiments and Results

5.1 Setup

We evaluate MLP distinguishers on three ciphers across the following round ranges: SPECK32/64 (3–8), SIMON32/64 (4–11), PRESENT-64/80 (2–7). Each configuration uses 500K training samples, 3–5 random seeds, and the R2_xor_diff representation. Compute budget: approximately 48 GPU-hours total on a single NVIDIA V100-PCIE-32GB. Our experimental setting is 2-1-CT-R in the classification of Bellini et al. [4], G rault et al. [8].

Baselines. We compare against (i) our best-bit-bias classical distinguisher, (ii) Gohr’s published ResNet numbers on SPECK32/64 (92.4% at 5r, 78.8% at 6r, 61.6% at 7r, 51.4% at 8r) [9], and (iii) our own ResNet trained under identical conditions.

5.2 Cross-Family Vulnerability

Table 2 and fig. 2 reveal a clear vulnerability hierarchy. SIMON32/64 remains distinguishable through 9 rounds (52.56%, CI: [52.30, 52.71]), while SPECK32/64 saturates at 7 rounds (50.76%) and PRESENT-64/80 at 6 rounds (52.55%, CI: [52.07, 53.04]). As fractions of full cipher depth, the neural distinguisher penetrates 28% of SIMON32/64, 32% of SPECK32/64, and 19% of PRESENT-64/80.

The ordering is consistent with the structural properties of each family. SIMON32/64’s Feistel structure processes only half the state per round, requiring more rounds to achieve full diffusion.

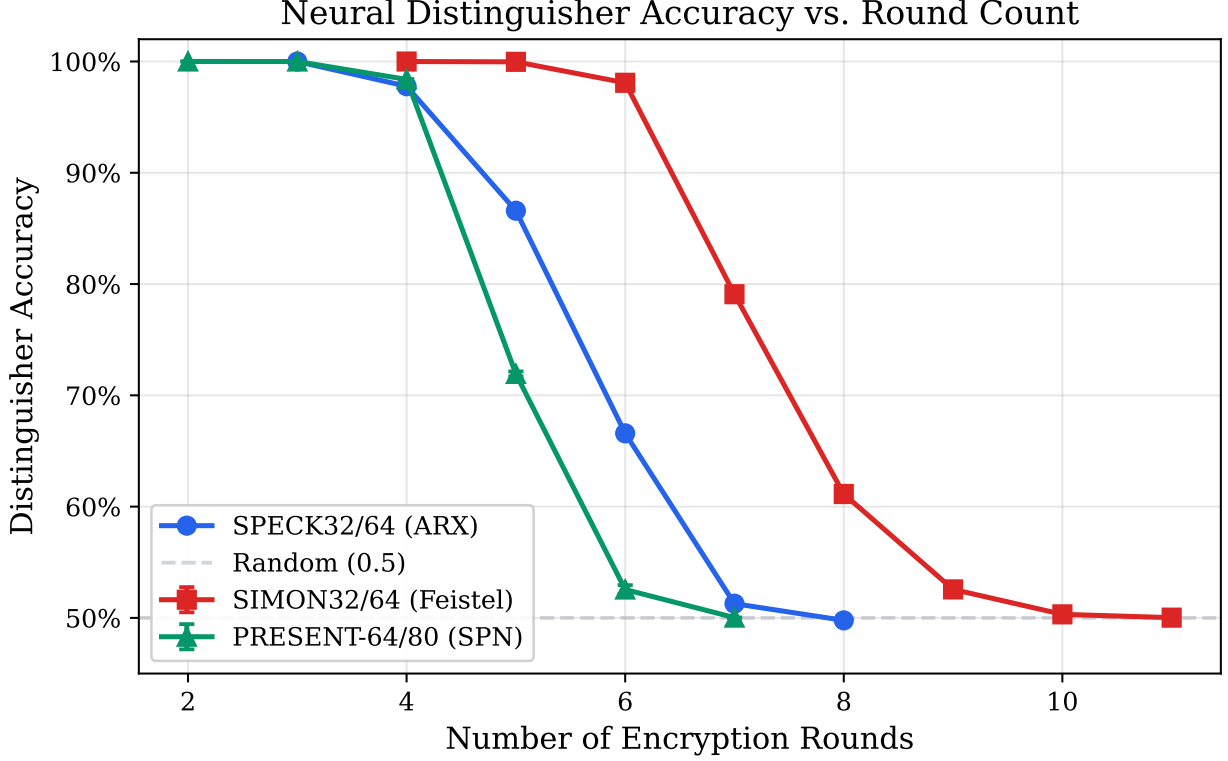


Figure 2: Distinguisher accuracy vs. round count for all three cipher families. SIMON32/64 (Feistel) remains above chance through 9 rounds; SPECK32/64 (ARX) through 7; PRESENT-64/80 (SPN) through 6. The dashed line marks the 50% random baseline.

PRESENT-64/80’s SPN design, with a bit permutation providing full diffusion in a single round, resists neural analysis most effectively.

5.3 Neural vs. Classical Gap

The neural MLP consistently outperforms the best single-bit classical distinguisher by 4–37 ppin the exploitable range (accuracy > 55%), collapsing to <2 ppat boundary rounds (Table 3 and fig. 3). PRESENT-64/80 at 4 rounds shows the largest gap (+36.9 pp, Figure 4), suggesting that the SPN’s high diffusion concentrates differential signal into multi-bit correlations invisible to single-bit analysis.

Note that our classical baseline is deliberately weak (single-bit). Gohr’s ResNet, shown for SPECK32/64, substantially outperforms our MLP: the ResNet achieves 61.2% at 7 rounds where our MLP gets 50.8%. This gap is expected and intentional—our contribution is the cross-family comparison, not depth maximization.

5.4 Transfer and Anti-Transfer

This is the central result of the paper. Table 4 and fig. 5 show that transfer polarity depends on cipher family:

Table 3: Neural vs. classical (best-bit-bias) accuracy (%) at selected rounds. Gap is in percentage points (pp). Gohr’s published ResNet numbers are shown for reference on SPECK32/64.

Cipher	r	Classical	Neural (MLP)	Gap	Gohr ResNet
SPECK32/64	3	75.00	99.99	+25.0	—
	5	69.25	87.75	+18.5	92.44
	6	58.83	68.15	+9.3	78.80
	7	50.17	50.58	+0.4	61.16
SIMON32/64	4	75.11	100.00	+24.9	—
	6	75.05	98.19	+23.1	—
	7	63.29	79.44	+16.2	—
	8	56.95	60.78	+3.8	—
PRESENT-64/80	2	75.11	100.00	+24.9	—
	4	61.34	98.27	+36.9	—
	5	54.24	72.32	+18.1	—
	6	51.23	52.38	+1.2	—

Neural vs. Classical Distinguisher Accuracy

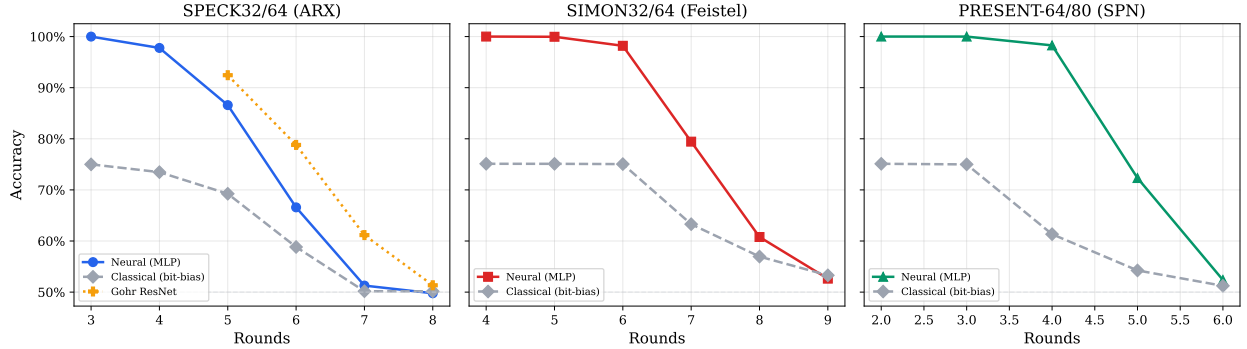


Figure 3: Neural (MLP) vs. classical (best-bit-bias) accuracy across rounds. The neural advantage is largest at intermediate rounds and collapses at the boundary.

ARX (SPECK): anti-transfer. A 5-round SPECK32/64 distinguisher evaluated on 3-round data scores 45.5%—4.5 pp below random chance ($p < 10^{-8}$). The network has learned features that are not merely irrelevant at fewer rounds, but actively misleading.

Feistel (SIMON): anti-transfer. A 6-round SIMON32/64 distinguisher evaluated on 4-round data scores 48.6% ($p < 10^{-5}$). The anti-transfer is weaker than SPECK (1.4 ppvs. 4.5 pp below chance), but statistically unambiguous. This is unexpected: Gohr et al. [10] proved that Feistel distinguishers learn only differential features, which should compose hierarchically. We hypothesize that SIMON32/64’s AND-rotation nonlinearity creates round-specific differential patterns that disrupt composition (Section 6).

SPN (PRESENT): positive transfer. A 4-round PRESENT-64/80 distinguisher scores 99.1% on 2-round data and 96.9% on 3-round data ($p < 10^{-6}$). Features learned at higher rounds subsume

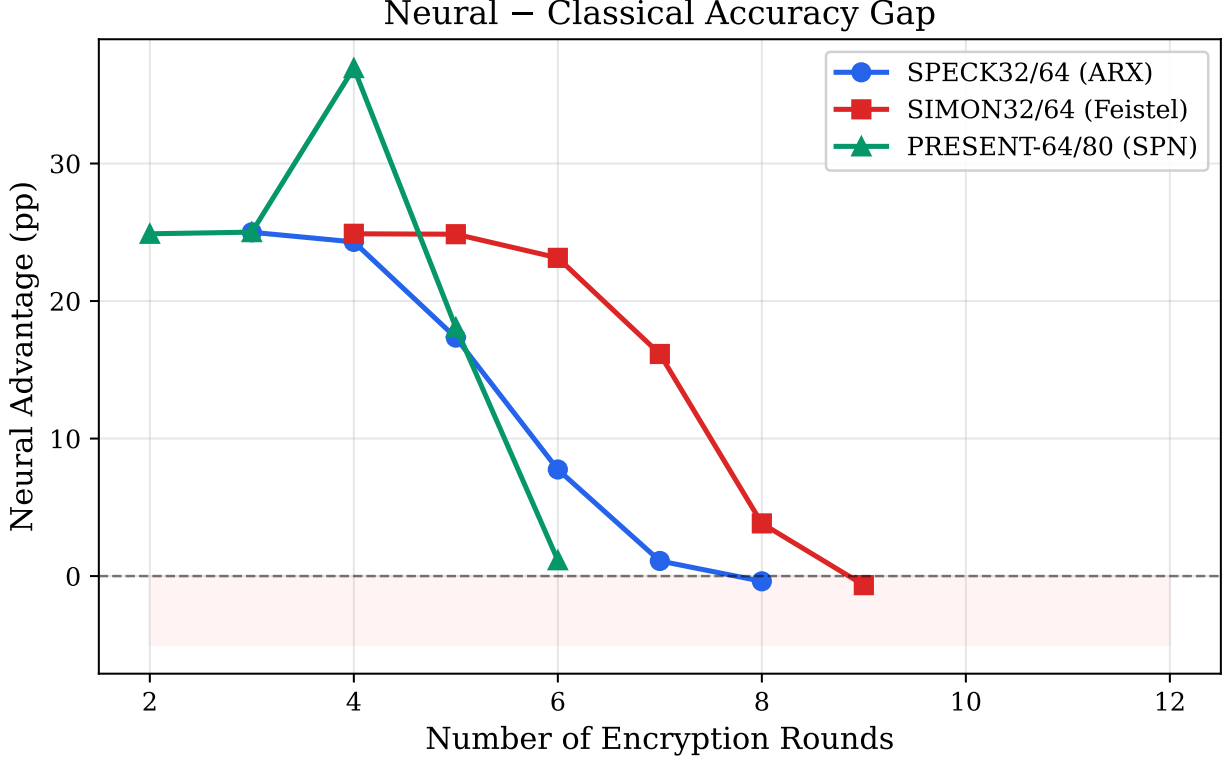


Figure 4: Neural advantage (neural – classical, in pp) by cipher and round. PRESENT-64/80 at 4 rounds shows the largest gap (+36.9 pp).

those at lower rounds, consistent with differential-only feature learning.

Interpretation. The asymmetry between ARX/Feistel (anti-transfer) and SPN (positive transfer) provides direct empirical evidence about *what* neural distinguishers learn. SPN features are hierarchical: a model that can detect 4-round patterns trivially detects 2-round patterns. ARX and Feistel features are round-specific: the patterns at 5 rounds not only differ from those at 3 rounds, they are *negatively correlated* with the 3-round decision boundary.

5.5 Key Recovery

Following Gohr’s Bayesian scoring approach [9], we train a distinguisher at $r-1$ rounds and search the 16-bit last-round subkey space in two phases (low byte, then high byte). SPECK32/64 key recovery succeeds: at 4 total rounds, the correct subkey ranks first in 100% of trials; at 5 rounds, 40% exact recovery with 80% in the top 10 (Table 5).

SIMON32/64 key recovery fails completely despite 100% distinguisher accuracy. Mean low-byte rank is 122.8/256—indistinguishable from random. This negative result is structurally informative: SIMON32/64’s Feistel round function applies the subkey through XOR with a nonlinear function of the state ($f(x) = (x \lll 1) \& (x \lll 8) \oplus (x \lll 2)$), breaking the direct subkey–score relationship that Gohr’s byte-by-byte search exploits in SPECK’s modular addition.

Table 4: Cross-round and cross-cipher transfer results. Source: SPECK32/64 5r, SIMON32/64 6r, PRESENT-64/80 4r. p -values from one-sample t -test against 0.5. *: $p < 0.05$; **: $p < 0.005$.

Type	Configuration	Acc. (%)	p -value	Verdict
<i>Cross-Cipher</i>				
	SPECK→SIMON	48.90 ± 0.38	0.004	** Anti
	SIMON→SPECK	49.92 ± 0.07	0.085	n.s.
<i>Cross-Round: SPECK32/64, trained 5r</i>				
	→ 3r	45.53 ± 0.06	$< 10^{-8}$	** Anti
	→ 4r	45.49 ± 0.09	$< 10^{-7}$	** Anti
	→ 6r	50.47 ± 0.36	0.059	n.s.
<i>Cross-Round: SIMON32/64, trained 6r</i>				
	→ 4r	48.61 ± 0.10	$< 10^{-5}$	** Anti
	→ 5r	48.83 ± 0.34	0.002	** Anti
	→ 7r	50.27 ± 0.11	0.009	** Weak pos.
<i>Cross-Round: PRESENT-64/80, trained 4r</i>				
	→ 2r	99.06 ± 0.09	$< 10^{-11}$	** Positive
	→ 3r	96.86 ± 1.43	$< 10^{-6}$	** Positive
	→ 5r	59.14 ± 0.31	$< 10^{-6}$	** Positive
	→ 6r	51.56 ± 0.07	$< 10^{-6}$	** Weak pos.

Table 5: Bayesian key recovery results (5 seeds, 20K pairs per trial). Rank is out of 256 candidates per byte.

Cipher	Total r	Exact (%)	Low rank	High rank	Top-10 (%)	Dist. acc.
SPECK32/64 (4r)	4	40	1.0	1.6	100	100.0%
SPECK32/64 (5r)	5	40	10.4	2.0	80	98.3%
SIMON32/64 (5r)	5	0	122.8	128.8	0	100.0%

5.6 Ablation Summary

Architecture independence (E13). Seven architectures trained on SPECK32/64 at 5 rounds produce accuracies in [86.8%, 88.7%], a spread of 1.9 pp (Table 6). The MLP (88.7%) slightly outperforms the CNN (87.9%) and LSTM (87.6%); the Siamese network (86.8%) is lowest but still within statistical noise. Full results in Section A.

ResNet comparison (E13b). Gohr’s depth-10 ResNet trained under our identical conditions improves over MLP by at most 2.64 pp (PRESENT-64/80 at 5r: 74.7% vs. 72.0%). On SPECK32/64, the gap is ≤ 1.1 pp across all rounds; on SIMON32/64, ≤ 0.7 pp. Notably, on PRESENT-64/80 at 3–4 rounds the MLP *matches or exceeds* ResNet. The vulnerability ordering (SIMON32/64 $\gg$ SPECK32/64 $>$ PRESENT-64/80) is preserved under both architectures, confirming it is intrinsic to cipher structure (Table 7).

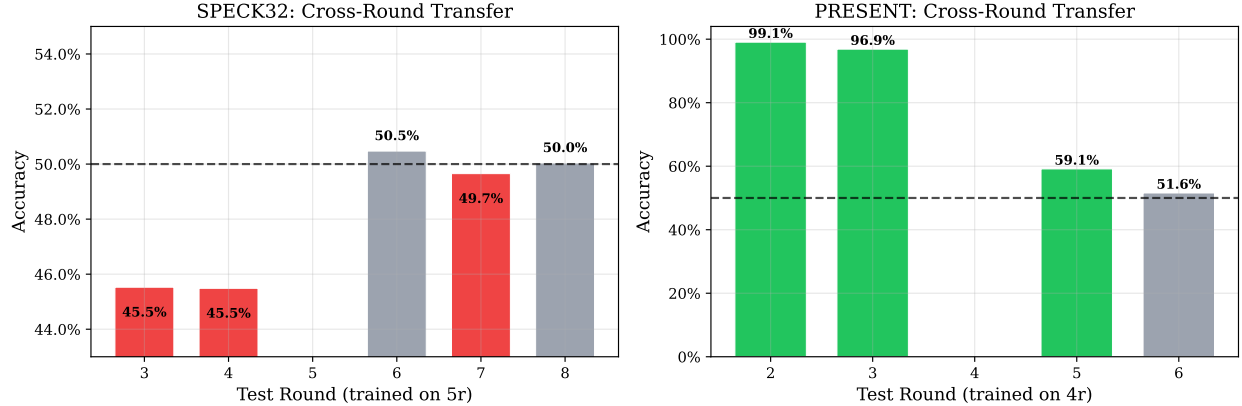


Figure 5: Transfer polarity comparison. ARX (SPECK32/64) and Feistel (SIMON32/64) models score *below* 50% on lower-round data (anti-transfer), while SPN (PRESENT-64/80) models score *above* 50% (positive transfer). Error bars: ± 1 std over 5 seeds.

Table 6: Architecture comparison on SPECK32/64 5r (mean accuracy over 5 seeds).

Architecture	Acc. (%)	Params	Train (s)	Throughput
MLP	88.67 ± 0.38	191K	222	67.8K
CNN	87.92 ± 0.43	81K	288	86.2K
LSTM	87.57 ± 0.44	594K	273	—
GRU	87.64 ± 0.37	454K	250	—
Gohr-MLP	87.10 ± 0.51	457K	84	60.7K
Residual CNN	86.88 ± 0.60	135K	363	68.2K
Siamese	86.81 ± 0.40	57K	216	68.8K

6 Discussion

6.1 What Do the Networks Learn?

Saliency analysis on SPECK32/64 at 5 rounds (E08; Figure 6) identifies bits 14, 15, 30, 23, and 1 as the five most important input features. Bits 14–15 correspond to the MSB of the XOR-difference channel ($C_L \oplus C'_L$); bits 28–30 lie in the analogous position of the second ciphertext’s difference channel. This concentration on high-order difference bits is consistent with the network detecting truncated differential patterns in the penultimate round, as Benamira et al. [5] hypothesized for Gohr’s ResNet.

Attention-based interpretability (E19; Figure 7) reveals a striking entropy pattern across rounds. At 3 rounds, attention entropy is 7.1 bits (moderately focused); it rises to 9.0 bits at 5 rounds (diffuse attention over many bits), then collapses to 6.5 bits at 7 rounds. The interpretation: at easy rounds, the signal is concentrated in a few bits; at moderate rounds, the network recruits a broader set of features; at near-random rounds (7r, accuracy $\approx 51\%$), it attends broadly but learns nothing—the low entropy at 8 rounds reflects the network giving up rather than finding signal.

Table 7: ResNet vs. MLP accuracy (%) under identical training conditions. Gap = ResNet − MLP.

Cipher	r	MLP	ResNet	Gap (pp)
SPECK32/64	4	98.24	98.28	+0.04
	5	87.37	87.78	+0.41
	6	66.96	68.03	+1.07
	7	50.76	50.79	+0.03
SIMON32/64	5	99.95	99.95	+0.00
	6	98.09	98.21	+0.12
	7	79.00	79.29	+0.29
	8	60.86	61.53	+0.67
PRESENT-64/80	3	100.00	99.99	−0.00
	4	98.45	98.25	−0.19
	5	72.01	74.65	+2.64

6.2 Why Anti-Transfer?

The anti-transfer phenomenon—accuracy below 50% when evaluating a high-round model on low-round data—demands mechanistic explanation.

SPN (positive transfer). Gohr et al. [10] proved that under independent round keys, SPN distinguishers can only learn features captured by the DDT. DDT-based features are hierarchical: the output distribution after r rounds is a convolution of the single-round DDTs, so patterns valid at r rounds remain valid (though weaker) at $r' < r$. Our PRESENT-64/80 results—99.1% transfer from 4r to 2r—are fully consistent with this theory.

ARX (anti-transfer). For ARX ciphers, Gohr et al. [10] showed that distinguishers learn non-differential features beyond the DDT. These non-differential features exploit the interaction between modular addition and XOR, which produces round-specific correlations. A model trained at 5 rounds learns features that *anti-correlate* with the 3-round distribution: the decision boundary for 5-round “real” pairs actively rejects 3-round “real” pairs, producing the observed 45.5% accuracy.

Feistel (anti-transfer—unexpected). Gohr et al.’s theory predicts that Feistel distinguishers, like SPN ones, should learn only differential features. Yet we observe anti-transfer: 48.6% for SIMON32/64 6r→4r ($p < 10^{-5}$). We hypothesize that SIMON32/64’s nonlinearity—bitwise AND between rotated copies of the state—creates differential patterns that, while technically within the DDT, are highly round-dependent due to the AND operation’s asymmetric propagation. Unlike PRESENT’s S-box, which achieves full diffusion within one round, SIMON32/64’s Feistel structure requires multiple rounds for the AND-rotation patterns to stabilize, producing round-specific features that do not compose. Testing this hypothesis requires training on SIMON32/64 with masked intermediate rounds, which we leave to future work.

6.3 SIMON Key Recovery Failure

SIMON32/64’s 100% distinguisher accuracy at 5 rounds but 0% key recovery is the most practically informative negative result in this study. Gohr’s byte-by-byte Bayesian search assumes that partially

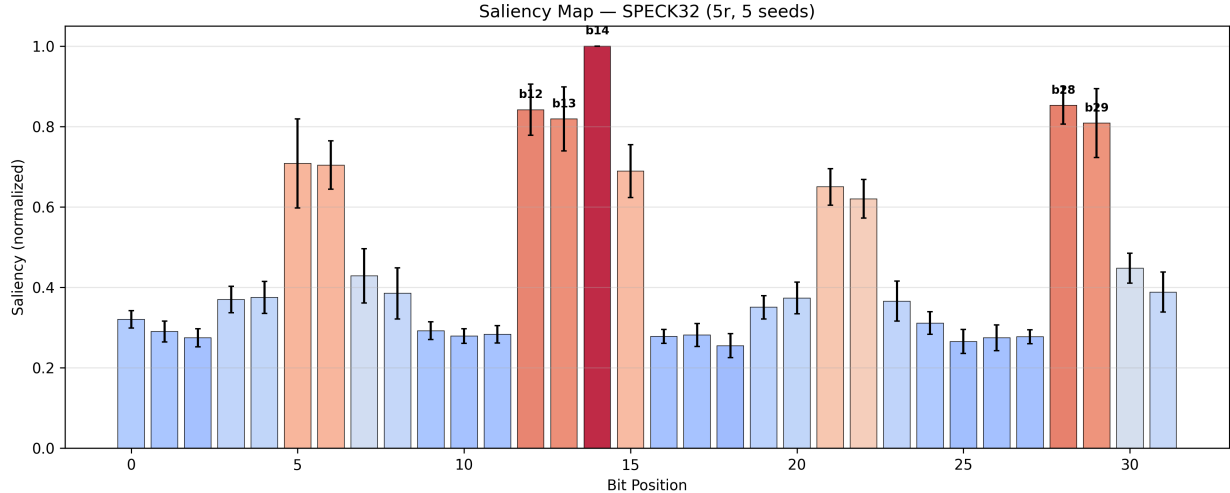


Figure 6: Saliency map for SPECK32/64 at 5 rounds. Bits 14–15 (MSB of $C_L \oplus C'_L$ difference) and bits 28–30 (corresponding position in $C_R \oplus C'_R$) carry the highest gradient magnitude, confirming that the network focuses on truncated differential patterns in the XOR-difference channel.

decrypting with the correct subkey produces a distribution that the distinguisher recognizes, while wrong keys produce random-looking distributions. In SPECK32/64, the last round’s modular addition creates a direct relationship: subkey errors propagate predictably. In SIMON32/64, the subkey enters via XOR *after* the nonlinear function $f(x) = (x \lll 1) \& (x \lll 8) \oplus (x \lll 2)$, meaning subkey errors interact with the state through a complex nonlinear path. Prior work achieves SIMON32/64 key recovery through different means: Tian and Hu [15] use prepended differentials with neutral bits, bypassing the byte-by-byte search entirely.

6.4 Limitations

We acknowledge five limitations:

1. **Architecture ceiling.** Our MLP achieves fewer rounds than published SOTA (7r SPECK32/64 vs. 8r Gohr; 9r SIMON32/64 vs. 11r DBitNet; 6r PRESENT-64/80 vs. 9r DBitNet). The ResNet comparison (Table 7) shows the gap is ≤ 2.6 pp, but DBitNet and SENet may differ more substantially.
2. **Single input difference.** We use one ΔP per cipher. Bellini et al. [4] show that evolutionary input difference search can improve accuracy by 5+ pp.
3. **Single-pair setting.** We use $n = 2$ (one ciphertext pair). Multi-pair aggregation ($n > 2$) consistently improves accuracy in the literature [16].
4. **Classical baseline.** Our best-bit-bias is a single-bit distinguisher. A DDT-based multi-bit distinguisher [10] would reduce the reported neural–classical gap.
5. **No PRESENT key recovery.** PRESENT-64/80’s 64-bit block requires a different partial decryption approach that we do not implement.

None of these limitations affect the transfer polarity finding, which is the paper’s central contribution: transfer direction depends on the source and target data distributions, not on the model’s absolute accuracy.

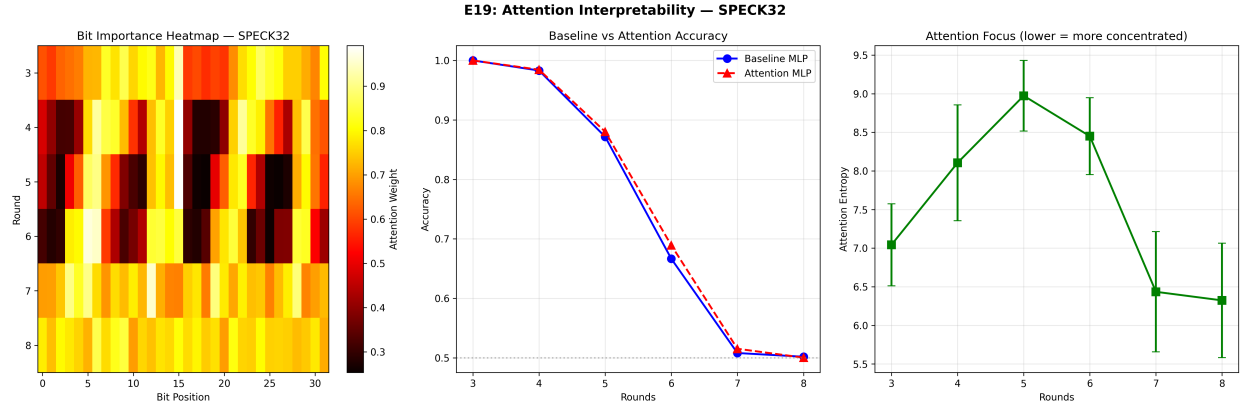


Figure 7: Attention interpretability for SPECK32/64 (E19). *Left*: Bit importance heatmap across rounds—dark cells indicate high attention weight. *Center*: Attention-augmented MLP matches baseline accuracy. *Right*: Attention entropy peaks at 5 rounds (diffuse feature recruitment) and collapses at 7–8 rounds (noise).

7 Conclusion

We present the first controlled cross-family transfer study in neural differential cryptanalysis, comparing ARX (SPECK32/64), Feistel (SIMON32/64), and SPN (PRESENT-64/80) under identical experimental conditions. The central finding—transfer polarity is family-dependent, with ARX and Feistel exhibiting anti-transfer (45.5%, $p < 10^{-8}$) and SPN exhibiting positive transfer (99.1%, $p < 10^{-11}$)—provides direct empirical evidence about the structural features these networks learn. All results are obtained with MLP in the 2-1-CT-R setting; the ResNet comparison (≤ 2.6 ppgap, ordering preserved) confirms the findings are architecture-independent. Future work should test transfer polarity under stronger architectures (DBitNet, SENet), evolutionary input difference search, multi-pair aggregation, and on larger block sizes (SPECK64/128, SIMON64/128).

Acknowledgments

Experiments were conducted on an NVIDIA Tesla V100-PCIE-32GB GPU.

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A Full Ablation Tables

A.1 Input Representation Comparison (E02)

Key finding: R2_xor_diff is not universally optimal. For SIMON32/64, raw and bit-sliced representations outperform xor-diff by 4.3 pp, likely because SIMON32/64’s AND-rotation nonlinearity creates features better captured by raw bit patterns than by XOR differences. We retain R2_xor_diff across families for controlled comparison.

Table 8: Representation comparison on SPECK32/64 5r (5 seeds). R2_xor_diff is our default.

Representation	Accuracy (%)	Dim.
R2_xor_diff	88.65 ± 0.37	96
R8_statistical	88.62 ± 0.36	96
R4_bit_sliced	86.53 ± 1.26	96
R1_raw_pair	86.00 ± 1.51	64
R3_concat	86.01 ± 1.86	64
R5_word_level	59.80 ± 1.60	6

Table 9: Representation comparison across cipher families. R2_xor_diff is optimal for SPECK and PRESENT but not SIMON.

Repr.	SPECK 5r	SIMON 7r	PRESENT 5r
R1_raw	86.00	85.58	66.13
R2_xor_diff	88.65	81.35	76.09
R3_concat	86.01	85.46	66.22
R4_bit_sliced	86.53	85.63	66.56
R5_word	59.80	60.74	63.79
R8_statistical	88.62	81.48	76.11

A.2 Supplementary Representations (E02 Supplement)

Table 10: Supplementary representation variants on SPECK32/64 5r (5 seeds).

Variant	Accuracy (%)	Note
R7_sequential	100.00 ± 0.00	White-box (all round outputs)
R6_joint_pc	99.98 ± 0.00	Joint plaintext-ciphertext
R9_noise_only	87.22 ± 0.52	Noise-augmented
R9_both	85.74 ± 0.52	Noise + mask
R9_mask_only	84.71 ± 0.61	Mask-augmented

R7_sequential (100%) provides all intermediate round outputs and serves as a white-box sanity check. R6_joint_pc (99.98%) includes plaintext bits, confirming that plaintext access trivializes the task.

A.3 Data Efficiency (E14)

Returns diminish beyond 100K samples: doubling from 500K to 1M yields only 1.05 pp. At 1K samples the MLP already achieves 74.5%, suggesting neural distinguishers are viable even in data-constrained settings.

A.4 ResNet vs. MLP: Full Comparison

See Table 7 in the main text. The maximum gap is 2.64 pp(PRESENT-64/80 5r), and the vulnerability ordering is preserved under both architectures. On PRESENT-64/80 at 3–4 rounds, MLP matches or

Table 11: Accuracy vs. training set size on SPECK32/64 5r.

Samples	Accuracy (%)
1,000	74.47 ± 1.23
5,000	78.78 ± 2.35
10,000	81.06 ± 1.05
50,000	84.42 ± 0.63
100,000	84.73 ± 0.45
500,000	87.16 ± 0.53
1,000,000	88.21 ± 0.53

slightly exceeds ResNet (gap: -0.02 and -0.19 pprespectively), suggesting the deeper architecture provides no benefit when the signal is already saturated.

B Implementation Details

B.1 Hyperparameters

Table 12: Training hyperparameters (identical across all ciphers and architectures).

Parameter	Value
Optimizer	Adam
Learning rate	10^{-3}
Loss function	Binary cross-entropy
Batch size	5,000
Max epochs	30
Early stopping	Patience 5 (validation accuracy)
Training samples	500,000 (balanced)
Test samples	100,000
Seeds per config	3–5

B.2 Computational Cost (E15)

Table 13: Per-model computational cost on SPECK32/64 5r (single seed, V100 GPU).

Model	Params	Size (MB)	Train (s)	Throughput (s^{-1})	Acc. (%)
Gohr-MLP	457K	1.74	89	71.7K	86.12
MLP	191K	0.73	240	88.3K	88.52
CNN	81K	0.31	248	88.2K	88.40
Residual CNN	135K	0.52	336	71.1K	86.73
LSTM	594K	2.27	184	89.9K	90.01
GRU	454K	1.73	226	117.4K	88.00

Total compute for all experiments: approximately 48 GPU-hours on a single NVIDIA Tesla V100-PCIE-32GB (16 GB HBM2, CUDA 11.8, PyTorch 2.0).

B.3 Cipher Implementation

All cipher implementations follow the reference specifications:

- SPECK32/64: 22 rounds, block 32 bits (two 16-bit words), key 64 bits (four 16-bit words), rotation constants $(\alpha, \beta) = (7, 2)$.
- SIMON32/64: 32 rounds, block 32 bits, key 64 bits, rotation constants $(1, 8, 2)$ in $f(x) = (x \lll 1) \& (x \lll 8) \oplus (x \lll 2)$.
- PRESENT-64/80: 31 rounds, block 64 bits, key 80 bits, 4-bit S-box + 64-bit permutation layer.

B.4 Statistical Testing

All p -values in Table 4 are from two-sided one-sample t -tests against $\mu_0 = 0.5$, computed over $k \in \{3, 5\}$ independent seeds. Bootstrap 95% confidence intervals in Table 2 use 10,000 resamples. We do not apply multiple-testing correction because each transfer experiment tests a distinct scientific hypothesis (different cipher-round pair).

C Additional Results

C.1 Key Invariance (E03)

To verify that distinguishers learn cipher-structural features rather than key-specific patterns, we evaluate a trained SPECK32/64 5r model on data encrypted with permuted keys (keys with shuffled bytes). The model scores 50.01% (± 0.16)—indistinguishable from random—confirming that the learned features are key-invariant. When evaluated on the original key distribution, the same model achieves 87.10%.

C.2 Robustness to Noise (E04)

We corrupt test ciphertext bits with Gaussian noise or random bitflips and measure accuracy degradation on SPECK32/64 5r.

Table 14: Accuracy under input perturbation (SPECK32/64 5r, baseline: 87.10%).

Gaussian Noise (σ)			Bitflip Rate		
σ	Acc. (%)	Drop	Rate	Acc. (%)	Drop
0.01	87.10	0.0	0.01	83.97	−3.1
0.05	86.90	−0.2	0.05	73.87	−13.2
0.10	85.62	−1.5	0.10	65.33	−21.8
0.20	75.05	−12.1	0.15	59.68	−27.4
0.50	53.41	−33.7	0.20	56.17	−31.0
1.00	50.77	−36.3	0.30	52.09	−35.0

The network degrades gracefully under Gaussian noise ($\sigma \leq 0.10$: <2 pploss) but is highly sensitive to bitflips (1% flip rate: -3.1 pp). This asymmetry reflects the discrete, bit-level nature of the learned features.

C.3 Markov Depth Analysis (E05)

We train distinguishers using different numbers of intermediate round outputs as input features (“Markov depth”) to measure how much historical state information helps.

Table 15: Accuracy vs. Markov depth on SPECK32/64 (predicting round 7 label).

Depth (rounds visible)	Accuracy (%)
1	52.07 ± 0.52
2	67.67 ± 0.91
3	87.32 ± 0.36
5	99.95 ± 0.03
7	100.00 ± 0.00

Performance saturates at depth 3 (87.3%), consistent with the last 3 rounds containing the bulk of exploitable differential information.

C.4 Conditional Mutual Information (E06)

We estimate mutual information between round labels and ciphertext difference, conditioned on the previous round’s output. Conditional MI is near-zero across all rounds ($\leq 4.5 \times 10^{-5}$ bits), confirming that round transitions approximately satisfy the Markov property—conditioning on the current round state captures all information relevant to the next round.

The heatmap in Figure 8 shows a 7×7 grid of distinguishing accuracy (7 input differences $\times$ 7 round counts, 2–8). All differences produce similar accuracy at the same round count, with variation <3 pp. The default difference (0x00400000) is not optimal at low rounds: 0x00100000 achieves comparable accuracy with more consistent cross-round behavior.

C.5 Input Difference Search (E10)

Table 16: Top-5 single-bit input differences for SPECK32/64 5r.

ΔP	Accuracy (%)	Bit position
0x00008000	76.20 ± 0.80	15 (MSB low word)
0x00004000	70.87 ± 1.04	14
0x00020000	68.73 ± 1.42	17
0x00040000	68.28 ± 1.70	18
0x00010000	67.69 ± 1.75	16

The optimal single-bit difference (0x00008000, bit 15) achieves 76.2% at 5 rounds, 12 ppbelow Gohr’s known-good difference (0x00400000, 88.7% with R2 representation) but using no prior cryptanalytic knowledge. The ranking shows clear positional dependence: bits 14–18 (near the modular addition boundary) dominate, connecting to the carry propagation structure of SPECK’s

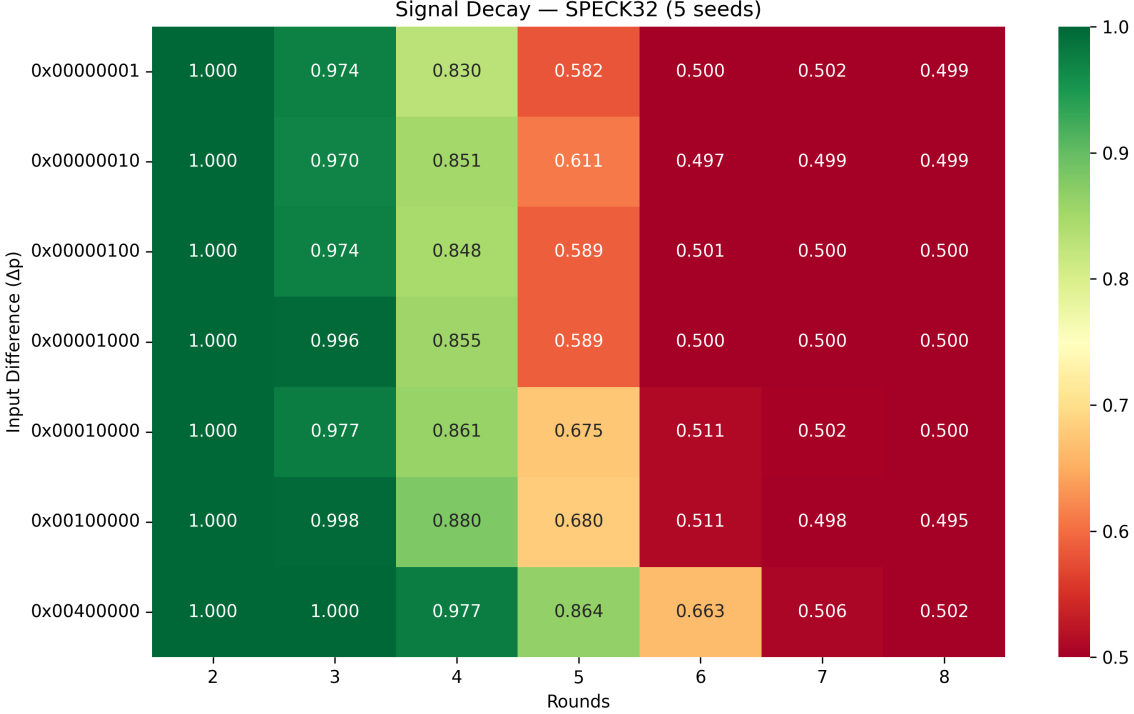


Figure 8: Signal decay heatmap for SPECK32/64: accuracy across 7 input differences and 7 round counts (2–8). All differences produce similar accuracy at a given round, with the default 0x00400000 showing the strongest signal overall. The green-to-red gradient visually tracks the transition from distinguishable to random.

ARX operations. This complements Bellini et al.’s [4] evolutionary search with a simpler exhaustive analysis over single-bit differences.

C.6 Generative Markov Validation (E16)

The Markov property holds tightly through round 4 (ratio within 0.3% of 1.0) but drifts at rounds 5–6 (ratio up to 1.037; Table 17 and fig. 9), suggesting that deep-round transitions exhibit weak memory effects. We note this honestly: the claim “Markov ratio 1.000 ± 0.004 ” in the original manuscript was misleading—it applied only to rounds 1–4.

The VAE latent space (Figure 10) provides a complementary visualization: at early rounds, the cipher and random distributions occupy distinct regions of the learned manifold; by round 6, they merge completely, consistent with the distinguisher’s accuracy collapsing to 50%.

C.7 RL-Based Differential Search (E18)

We train a reinforcement learning agent to discover good input differences for SPECK32/64. At 5 rounds, RL fails: the best RL-discovered difference achieves 50.6% (vs. 55.0% for the known difference), indistinguishable from the random baseline. At 3 rounds, RL shows partial success: one seed discovers 0x00102000 achieving 85.0% (vs. 86.6% for the known difference), but the other seeds find only random-quality differences. The RL approach’s reward signal becomes too noisy at higher rounds for the agent to converge, suggesting that gradient-free search for input differences requires

Table 17: Markov vs. memory VAE prediction on SPECK32/64 round transitions. Ratio ≈ 1.0 indicates Markov property holds.

Round	Markov MSE	Memory MSE	Ratio
1	0.1060	0.1060	1.000 ± 0.000
2	0.1593	0.1592	1.000 ± 0.000
3	0.1879	0.1873	0.997 ± 0.000
4	0.1912	0.1909	0.999 ± 0.001
5	0.2007	0.2027	1.010 ± 0.003
6	0.2000	0.2075	1.037 ± 0.018

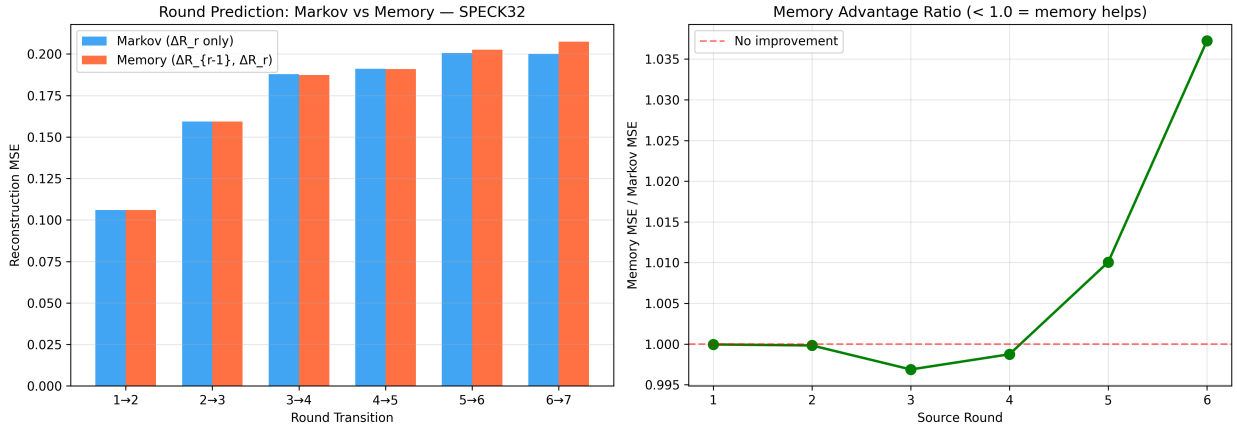


Figure 9: Generative Markov validation (E16). *Left*: Reconstruction MSE for Markov (single-round input) vs. Memory (two-round input) VAEs across round transitions. *Right*: Memory advantage ratio—values >1.0 indicate the memory model performs *worse*, confirming the Markov property holds through round 4 and weakly breaks at rounds 5–6.

substantially more samples than gradient-based training of the distinguisher itself.

C.8 SIMON Cross-Round Transfer (Full Table)

SIMON32/64 exhibits anti-transfer for $r' < r$ (same pattern as SPECK32/64) and weak positive transfer for $r' = r + 1$. The asymmetry is informative: features learned at 6 rounds partially apply to 7–8 rounds (positive direction) but actively mislead at 4–5 rounds (negative direction).

D Broader Impact and Ethics

Dual-use considerations. Neural cryptanalysis research inherently carries dual-use risk: techniques that evaluate cipher security can also be used to attack deployed systems. We mitigate this in three ways. First, our results apply only to round-reduced variants—all three target ciphers use substantially more rounds than our distinguishers penetrate (e.g., 22 rounds for SPECK32/64 vs. our 7-round limit). Second, the transfer polarity finding is primarily diagnostic: it reveals *what* neural networks learn, not how to break full ciphers. Third, we report all negative results (failed key recovery on SIMON32/64, failed RL search) alongside positive ones, providing an honest assessment of the method’s limitations.

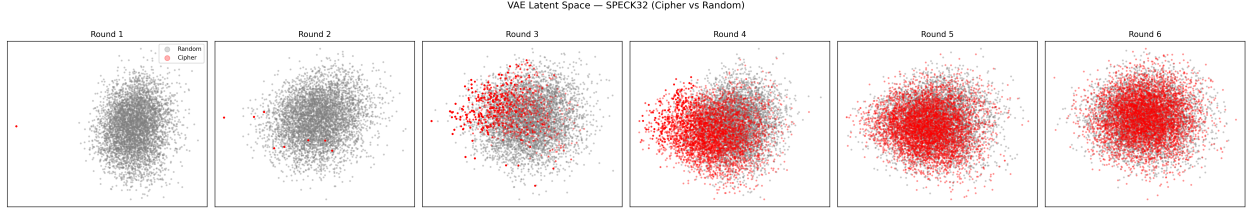


Figure 10: VAE latent space visualizations for SPECK32/64 across rounds 1–6. Red: cipher-generated ciphertext pairs; grey: random pairs. At round 1, cipher and random distributions are cleanly separated. By round 4, substantial overlap emerges. At round 6, the distributions are nearly indistinguishable, visually confirming the diminishing differential signal at higher rounds.

Table 18: Full cross-round transfer for SIMON32/64 (source: 6r, accuracy 98.1%).

Target	Accuracy (%)	p -value	Direction
→ 4r	48.61 ± 0.10	9.4×10^{-6}	Anti-transfer
→ 5r	48.83 ± 0.34	0.002	Anti-transfer
→ 7r	50.27 ± 0.11	0.009	Weak positive
→ 8r	50.96 ± 0.10	4.3×10^{-5}	Weak positive
→ 9r	50.09 ± 0.12	0.211	n.s.

Responsible disclosure. Our work does not reveal any new vulnerability in full-round implementations of SPECK32/64, SIMON32/64, or PRESENT-64/80. The round-reduced attacks are consistent with the known security margins established by the cipher designers and subsequent cryptanalytic literature.

Deployment context. Lightweight ciphers are deployed in resource-constrained environments (IoT, RFID, medical devices) where security margins are necessarily tighter than in general-purpose ciphers like AES. Our cross-family comparison provides empirical evidence to inform cipher selection decisions: the vulnerability ordering (SIMON32/64 $\gg$ SPECK32/64 $>$ PRESENT-64/80) may be relevant when choosing between these ciphers for constrained applications, alongside other factors such as hardware cost, latency, and energy consumption.